

Markscheme

Mathematics: analysis and approaches
Practice question bank

149. (a)

Triangle ABC on an Argand diagram has vertices $z_A = 0$ and $z_B = 6$, and is equilateral with $\text{Im}(z_C) > 0$.

Find z_C , obtained by rotating z_B by 60° anticlockwise about z_A , in the form $a + bi$ correct to 3 significant figures.

$$z_C = z_B \times (\cos 60^\circ + i \sin 60^\circ) \quad (M1)$$

$$= 6(0.5 + 0.866i) \quad (M1)$$

$$= 3 + 5.20i \quad \mathbf{A1}$$

[6 marks]